

Department of Applied Mathematics and Theoretical Physics
Visiting Graduate Student with Blake D. Sherwin

May - Jul. 2020
Cambridge, UK

Max-Planck-Institut für Astronomie
Summer Intern with Jan Rybizki

Jul. - Sep. 2017
Heidelberg, Germany

Center for Astrophysics | Harvard & Smithsonian
Undergraduate Research Fellow with Ákos Bogdán

Jun. - Aug. 2016
Cambridge, USA

PUBLICATION LIST

* = Author list alphabetized

Major Author

1. Smith, A. F., Ivanov, M. M., de Belsunce, R., Chen, S.-F., Chudaykin, A., **Philcox, O. H. E.**, “ $w_0 w_a$ or bc : DESI constraints on relative baryon-CDM perturbations”, ([arXiv](#)).
2. *Chudaykin, A., Ivanov, M. M., Kallosh, R., Linde, A., **Philcox, O. H. E.**, Yamada, Y., “Inflation, Open Universes, and Dark Energy”, ([arXiv](#)).
3. Jacobus, C., Tucci, B., **Philcox, O. H. E.**, “Emulating Cosmic Structure Formation with a Lagrangian Neural Cellular Automaton”, ([arXiv](#)).
4. **Philcox, O. H. E.**, “Dissecting the Scalar Cosmological Collider with the Cosmic Microwave Background”, ([arXiv](#)).
5. Yang, S.-X., **Philcox, O. H. E.**, “Searching for Folded Primordial Non-Gaussianity with Galaxy Surveys”, ([arXiv](#)).
6. **Philcox, O. H. E.**, “The Limitations of Cosmological Collider Analyses”, ([arXiv](#)).
7. Jense, H. T., El-Haj, L. H. A., Hill, J. C., **Philcox, O. H. E.**, “Probing inflationary particle production with the CMB power spectrum”, ([arXiv](#)).
8. *Gallardo, P., Pardo, K., **Philcox, O. H. E.**, Battaglia, N., Battistelli, E., Bean, R., *et al.*, “Test of the Gravitational Force Law on Cosmological Scales Using the Kinematic Sunyaev-Zeldovich Effect”, *Phys. Rev. Lett.* **136**, 151002 (2026) ([arXiv](#)).
9. *Chen, S.-F., DeRose, J., Ivanov, M. M., **Philcox, O. H. E.**, “Cosmic Shear in Effective Field Theory at Two-Loop Order: Revisiting S_8 in Dark Energy Survey Data”, ([arXiv](#)).
10. **Philcox, O. H. E.**, “What Shape is the Inflationary Bispectrum?”, ([arXiv](#)).
11. ***Philcox, O. H. E.**, Pimentel, G. L., Yang, C., “Searching for unparticles with the cosmic microwave background”, *Phys. Rev. D* **113**, 123529 (2026) ([arXiv](#)).
12. *Chudaykin, A., Ivanov, M. M., **Philcox, O. H. E.**, “Reanalyzing DESI DR1: 5. Cosmological Constraints with Simulation-Based Priors”, ([arXiv](#)).
13. Ivanov, M. M., Sullivan, J. M., de Belsunce, R., Chen, S.-F., Chudaykin, A., Maus, M., *et al.* (inc. **Philcox, O. H. E.**), “Reanalyzing DESI DR1: 4. Percent-Level Cosmological Constraints from Combined Probes and Robust Evidence for the Normal Neutrino Mass Hierarchy”, *submitted to Phys. Rev. D* ([arXiv](#)).
14. *Chudaykin, A., Ivanov, M. M., **Philcox, O. H. E.**, “Reanalyzing DESI DR1. III. Constraints on inflation from galaxy power spectra and bispectra”, *Phys. Rev. D* **113**, 063552 (2026) ([arXiv](#)).
15. *Chudaykin, A., Ivanov, M. M., **Philcox, O. H. E.**, “Reanalyzing DESI DR1. II. Constraints on dark energy, spatial curvature, and neutrino masses”, *Phys. Rev. D* **113**, 123506 (2026) ([arXiv](#)).
16. **Philcox, O. H. E.**, Zhong, K., Sirletti, S. S., “Separating the Inseparable: Constraining Arbitrary Primordial Bispectra with Cosmic Microwave Background Data”, *submitted to Phys. Rev. D* ([arXiv](#)).
17. El-Haj, L. H. A., **Philcox, O. H. E.**, Hill, J. C., “Constraining inflationary particle production with CMB polarization”, *Phys. Rev. D* **113**, 103518 (2026) ([arXiv](#)).

18. *Bakx, T., Ivanov, M. M., **Philcox, O. H. E.**, Vlah, Z., “One-Loop Galaxy Bispectrum: Consistent Theory, Efficient Analysis with COBRA, and Implications for Cosmological Parameters”, *submitted to Phys. Rev. D* ([arXiv](#)).
19. *Chudaykin, A., Ivanov, M. M., **Philcox, O. H. E.**, “Reanalyzing DESI DR1. I. Λ CDM constraints from the power spectrum and bispectrum”, *Phys. Rev. D* **113**, 063502 (2026) ([arXiv](#)).
20. ***Philcox, O. H. E.**, Silverstein, E., Torroba, G., “Quantum stress-energy at timelike boundaries: Testing a new beyond- Λ CDM parameter with cosmological data”, *Phys. Rev. D* **113**, 043548 (2026) ([arXiv](#)).
21. **Philcox, O. H. E.**, Hill, J. C., “Integrated Sachs-Wolfe lensing bispectrum and trispectrum”, *Phys. Rev. D* **111**, 123554 (2025) ([arXiv](#)).
22. Goldstein, S., **Philcox, O. H. E.**, Fondi, E., Coulton, W. R., “Wonderings on wiggly bispectra: Nonlinear evolution and reconstruction of oscillations in the squeezed bispectrum”, *Phys. Rev. D* **112**, 083503 (2025) ([arXiv](#)).
23. Zhang, G., Modi, C., **Philcox, O. H. E.**, “Modeling galaxy surveys with hybrid simulation-based inference”, *Phys. Rev. D* **113**, 043518 (2026) ([arXiv](#)).
24. **Philcox, O. H. E.**, “Searching for inflationary physics with the CMB trispectrum. III. Constraints from *Planck*”, *Phys. Rev. D* **111**, 123534 (2025) ([arXiv](#)).
25. **Philcox, O. H. E.**, “Searching for inflationary physics with the CMB trispectrum. II. Code and validation”, *Phys. Rev. D* **111**, 123533 (2025) ([arXiv](#)).
26. **Philcox, O. H. E.**, “Searching for inflationary physics with the CMB trispectrum. I. Primordial theory and optimal estimators”, *Phys. Rev. D* **111**, 123532 (2025) ([arXiv](#)).
27. ***Philcox, O. H. E.**, Shiraishi, M., “Non-Gaussianity beyond the scalar sector: A search for tensor and mixed tensor-scalar bispectra with Planck data”, *Phys. Rev. D* **111**, 123502 (2025) ([arXiv](#)).
28. Goldstein, S., **Philcox, O. H. E.**, Hill, J. C., Hui, L., “Intermediate mass-range particles from small scales: Nonperturbative techniques for cosmological collider physics from large-scale structure surveys”, *Phys. Rev. D* **110**, 083516 (2024) ([arXiv](#)).
29. *Coulton, W. R., **Philcox, O. H. E.**, Villaescusa-Navarro, F., “The Impact of Non-Gaussian Primordial Tails on Cosmological Observables”, *submitted to Phys. Rev. D* ([arXiv](#)).
30. *Chen, S.-F., Ivanov, M. M., **Philcox, O. H. E.**, Wenzl, L., “Suppression without Thawing: Constraining Structure Formation and Dark Energy with Galaxy Clustering”, *Phys. Rev. Lett.* **133**, 231001 (2024) ([arXiv](#)).
31. **Philcox, O. H. E.**, Kumar, S., Hill, J. C., “Searching for inflationary particle production in Planck data”, *Phys. Rev. D* **111**, 103523 (2025) ([arXiv](#)).
32. **Philcox, O. H. E.**, Flöss, T., “Suite of optimal and efficient power spectrum and bispectrum estimators for large-scale structure analyses”, *Phys. Rev. D* **112**, 063507 (2025) ([arXiv](#)).
33. Cabass, G., **Philcox, O. H. E.**, Ivanov, M. M., Akitsu, K., Chen, S.-F., Simonović, M., *et al.*, “BOSS constraints on massive particles during inflation: The cosmological collider in action”, *Phys. Rev. D* **111**, 063510 (2025) ([arXiv](#)).
34. de Belsunce, R., **Philcox, O. H. E.**, Irsic, V., McDonald, P., Guy, J., Palanque-Delabrouille, N., “The 3D Lyman- α forest power spectrum from eBOSS DR16”, *MNRAS* **533**, 3756 – 3770 (2024) ([arXiv](#)).
35. **Philcox, O. H. E.**, Ereza, J., “Could Sample Variance be Responsible for the Parity-Violating Signal Seen in the BOSS Galaxy Survey?”, *Phil. Trans. Roy. Soc. A* **383**, 2290 (2025) ([arXiv](#)).
36. ***Philcox, O. H. E.**, Shiraishi, M., “Testing graviton parity and Gaussianity with Planck T -, E -, and B -mode bispectra”, *Phys. Rev. D* **109**, 063522 (2024) ([arXiv](#)).
37. Goldstein, S., **Philcox, O. H. E.**, Hill, J. C., Esposito, A., Hui, L., “Consistently constraining f_{NL} with the squeezed lensing bispectrum using consistency relations”, *Phys. Rev. D* **109**, 043515 (2024) ([arXiv](#)).
38. *Modi, C., **Philcox, O. H. E.**, “Hybrid SBI or How I Learned to Stop Worrying and Learn the Likelihood”, *submitted to Phys. Rev. Lett.* ([arXiv](#)).

39. **Philcox, O. H. E.**, König, M. J., Alexander, S., Spergel, D. N., “What can galaxy shapes tell us about physics beyond the standard model?”, *Phys. Rev. D* **109**, 063541 (2024) ([arXiv](#)).
40. ***Philcox, O. H. E.**, Shiraishi, M., “Testing parity symmetry with the polarized cosmic microwave background”, *Phys. Rev. D* **109**, 083514 (2024) ([arXiv](#)).
41. *Coulton, W. R., **Philcox, O. H. E.**, Villaescusa-Navarro, F., “Signatures of a parity-violating universe”, *Phys. Rev. D* **109**, 023531 (2024) ([arXiv](#)).
42. **Philcox, O. H. E.**, “Optimal estimation of the binned mask-free power spectrum, bispectrum, and trispectrum on the full sky: Tensor edition”, *Phys. Rev. D* **108**, 063506 (2023) ([arXiv](#)).
43. *Ivanov, M. M., **Philcox, O. H. E.**, “Measuring H_0 with Spectroscopic Surveys”, chapter in “Hubble Constant Tension” (Eds. Di Valentino, E. and Brout, D., *Springer*, Singapore) ([arXiv](#)).
44. **Philcox, O. H. E.**, “Do the CMB Temperature Fluctuations Conserve Parity?”, *Phys. Rev. Lett.* **131**, 181001 (2023) ([arXiv](#)).
45. **Philcox, O. H. E.**, “Optimal estimation of the binned mask-free power spectrum, bispectrum, and trispectrum on the full sky: Scalar edition”, *Phys. Rev. D* **107**, 123516 (2023) ([arXiv](#)).
46. Creque-Sarbinowski, C., Alexander, S., Kamionkowski, M., **Philcox, O. H. E.**, “Parity-violating trispectrum from Chern-Simons gravity”, *JCAP* **11** 029 (2023) ([arXiv](#)).
47. Surrao, K. M., **Philcox, O. H. E.**, Hill, J. C., “Accurate estimation of angular power spectra for maps with correlated masks”, *Phys. Rev. D* **107**, 083521 (2023) ([arXiv](#)).
48. Ivanov, M. M., **Philcox, O. H. E.**, Cabass, G., Nishimichi, T., Simonović, M., Zaldarriaga, M., “Cosmology with the galaxy bispectrum multipoles: Optimal estimation and application to BOSS data”, *Phys. Rev. D* **107**, 083515 (2023) ([arXiv](#)).
49. *Cabass, G., Ivanov, M. M., **Philcox, O. H. E.**, Simonovic, M., Zaldarriaga, M., “Constraining single-field inflation with MegaMapper”, *Phys. Lett. B* **841**, 137912 (2023) ([arXiv](#)).
50. *Cabass, G., Ivanov, M. M., **Philcox, O. H. E.**, “Colliders and ghosts: Constraining inflation with the parity-odd galaxy four-point function”, *Phys. Rev. D* **107**, 023523 (2023) ([arXiv](#)).
51. Goldstein, S., Esposito, A., **Philcox, O. H. E.**, Hui, L., Hill, J. C., Scoccimarro, R., *et al.*, “Squeezing f_{NL} out of the matter bispectrum with consistency relations”, *Phys. Rev. D* **106**, 123525 (2022) ([arXiv](#)).
52. ***Philcox, O. H. E.**, Torquato, S., “Disordered Heterogeneous Universe: Galaxy Distribution and Clustering across Length Scales”, *Phys. Rev. X* **13**, 011038 (2023) ([arXiv](#)).
53. **Philcox, O. H. E.**, Johnson, M. C., “Novel cosmological tests from combining galaxy lensing and the polarized Sunyaev-Zel’dovich effect”, *Phys. Rev. D* **106**, 083501 (2022) ([arXiv](#)).
54. **Philcox, O. H. E.**, “Probing parity violation with the four-point correlation function of BOSS galaxies”, *Phys. Rev. D* **106**, 063501 (2022) ([arXiv](#)).
55. **Philcox, O. H. E.**, Ivanov, M. M., Cabass, G., Simonović, M., Zaldarriaga, M., Nishimichi, T., “Cosmology with the redshift-space galaxy bispectrum monopole at one-loop order”, *Phys. Rev. D* **106**, 043530 (2022) ([arXiv](#)).
56. **Philcox, O. H. E.**, Farren, G. S., Sherwin, B. D., Baxter, E. J., Brout, D. J., “Determining the Hubble constant without the sound horizon: A 3.6% constraint on H_0 from galaxy surveys, CMB lensing, and supernovae”, *Phys. Rev. D* **106**, 063530 (2022) ([arXiv](#)).
57. *Cabass, G., Ivanov, M. M., **Philcox, O. H. E.**, Simonović, M., Zaldarriaga, M., “Constraints on multifield inflation from the BOSS galaxy survey”, *Phys. Rev. D* **106**, 043506 (2022) ([arXiv](#)).
58. *Cabass, G., Ivanov, M. M., **Philcox, O. H. E.**, Simonović, M., Zaldarriaga, M., “Constraints on Single-Field Inflation from the BOSS Galaxy Survey”, *Phys. Rev. Lett.* **129**, 021301 (2022) ([arXiv](#)).
59. *Farren, G. S., **Philcox, O. H. E.**, Sherwin, B. D., “Determining the Hubble constant without the sound horizon: Perspectives with future galaxy surveys”, *Phys. Rev. D* **105**, 063503 (2022) ([arXiv](#)).
60. **Philcox, O. H. E.**, Ivanov, M. M., “BOSS DR12 full-shape cosmology: Λ CDM constraints from the large-scale galaxy power spectrum and bispectrum monopole”, *Phys. Rev. D* **105**, 043517 (2022) ([arXiv](#)).

61. Ivanov, M. M., **Philcox, O. H. E.**, Nishimichi, T., Simonović, M., Takada, M., Zaldarriaga, M., “Precision analysis of the redshift-space galaxy bispectrum”, *Phys. Rev. D* **105**, 063512 (2022) ([arXiv](#)).
62. Ivanov, M. M., **Philcox, O. H. E.**, Simonović, M., Zaldarriaga, M., Nishimichi, T., Takada, M., “Cosmological constraints without nonlinear redshift-space distortions”, *Phys. Rev. D* **105**, 043531 (2022) ([arXiv](#)).
63. **Philcox, O. H. E.**, Hou, J., Slepian, Z., “A First Detection of the Connected 4-Point Correlation Function of Galaxies Using the BOSS CMASS Sample”, *submitted to Phys. Rev. D* ([arXiv](#)).
64. **Philcox, O. H. E.**, “Cosmology without window functions. II. Cubic estimators for the galaxy bispectrum”, *Phys. Rev. D* **104**, 123529 (2021) ([arXiv](#)).
65. ***Philcox, O. H. E.**, Slepian, Z., “Efficient computation of N-point correlation functions in D dimensions”, *PNAS* **119**, 33 (2022) ([arXiv](#)).
66. **Philcox, O. H. E.**, Slepian, Z., Hou, J., Warner, C., Cahn, R. N., Eisenstein, D. J., “encore: an O (Ng²) estimator for galaxy N-point correlation functions”, *MNRAS* **509**, 2457 – 2481 (2021) ([arXiv](#)).
67. ***Philcox, O. H. E.**, Slepian, Z., “An exact integral-to-sum relation for products of Bessel functions”, *Proc. Roy. Soc. A* **477**, 2253 (2021) ([arXiv](#)).
68. Slepian, Z., **Philcox, O. H. E.**, “A uniform spherical goat (problem): explicit solution for homologous collapse’s radial evolution in time”, *MNRAS* **522**, L42-L45 (2023) ([arXiv](#)).
69. ***Philcox, O. H. E.**, Slepian, Z., “Beyond the Yamamoto approximation: Anisotropic power spectra and correlation functions with pairwise lines of sight”, *Phys. Rev. D* **103**, 123509 (2021) ([arXiv](#)).
70. **Philcox, O. H. E.**, “Cosmology without window functions: Quadratic estimators for the galaxy power spectrum”, *Phys. Rev. D* **103**, 103504 (2021) ([arXiv](#)).
71. **Philcox, O. H. E.**, Aviles, A., Massara, E., “Modeling the Marked Spectrum of Matter and Biased Tracers in Real- and Redshift-Space”, *JCAP* **03** 038 (2021) ([arXiv](#)).
72. **Philcox, O. H. E.**, Ivanov, M. M., Zaldarriaga, M., Simonovic, M., Schmittfull, M., “Fewer Mocks and Less Noise: Reducing the Dimensionality of Cosmological Observables with Subspace Projections”, *Phys. Rev. D* **103**, 043508 (2021) ([arXiv](#)).
73. **Philcox, O. H. E.**, Sherwin, B. D., Farren, G. S., Baxter, E. J., “Determining the Hubble Constant without the Sound Horizon: Measurements from Galaxy Surveys”, *Phys. Rev. D* **103**, 023538 (2021) ([arXiv](#)).
74. **Philcox, O. H. E.**, Massara, E., Spergel, D. N., “What does the marked power spectrum measure? Insights from perturbation theory”, *Phys. Rev. D* **102**, 043516 (2020) ([arXiv](#)).
75. **Philcox, O. H. E.**, “A faster Fourier transform? Computing small-scale power spectra and bispectra for cosmological simulations in $\mathcal{O}(N^2)$ time”, *MNRAS* **501**, 4004 – 4034 (2021) ([arXiv](#)).
76. ***Philcox, O. H. E.**, Spergel, D. N., Villaescusa-Navarro, F., “Effective halo model: Creating a physical and accurate model of the matter power spectrum and cluster counts”, *Phys. Rev. D* **101**, 123520 (2020) ([arXiv](#)).
77. **Philcox, O. H. E.**, Ivanov, M. M., Simonović, M., Zaldarriaga, M., “Combining Full-Shape and BAO Analyses of Galaxy Power Spectra: A 1.6% CMB-independent constraint on H_0 ”, *JCAP* **05** 032 (2020) ([arXiv](#)).
78. **Philcox, O. H. E.**, Eisenstein, D. J., “Computing the Small-Scale Galaxy Power Spectrum and Bispectrum in Configuration-Space”, *MNRAS* **492**, 1214 – 1242 (2020) ([arXiv](#)).
79. **Philcox, O. H. E.**, Eisenstein, D. J., “Estimating Covariance Matrices for Two- and Three-Point Correlation Function Moments in Arbitrary Survey Geometries”, *MNRAS* **490**, 5931 – 5951 (2019) ([arXiv](#)).
80. ***Philcox, O. H. E.**, Rybizki, J., “Inferring Galactic Parameters from Chemical Abundances: A Multi-Star Approach”, *ApJ* **887**, 9 (2019) ([arXiv](#)).
81. **Philcox, O. H. E.**, Eisenstein, D. J., O’Connell, R., Wiegand, A., “rascalc: a jackknife approach to estimating single- and multitracers galaxy covariance matrices”, *MNRAS* **491**, 3290 – 3317 (2020) ([arXiv](#)).
82. ***Philcox, O. H. E.**, Sherwin, B. D., van Engelen, A., “Detection and Removal of B-mode Dust Foregrounds with Signatures of Statistical Anisotropy”, *MNRAS* **479**, 5577 – 5595 (2018) ([arXiv](#)).

83. **Philcox, O. H. E.**, Rybizki, J., Gutcke, T., “On the Optimal Choice of Nucleosynthetic Yields, Initial Mass Function, and Number of SNe Ia for Chemical Evolution Modeling”, *ApJ* **861**, 40 (2018) ([arXiv](#)).

Contributing Author

84. *Besuner, R., Dey, A., Drlica-Wagner, A., Ebina, H., Fernandez Moroni, G., Ferraro, S., *et al.* (inc. **Philcox, O. H. E.**), “The Spectroscopic Stage-5 Experiment”, *White Paper* ([arXiv](#)).
85. *Krause, E., Kobayashi, Y., Salcedo, A. N., Ivanov, M. M., Abel, T., Akitsu, K., *et al.* (inc. **Philcox, O. H. E.**), “A Parameter-Masked Mock Data Challenge for Beyond-Two-Point Galaxy Clustering Statistics”, *ApJ* **990**, 99 (2025) ([arXiv](#)).
86. Rogers, K. K., Hložek, R., Laguë, A., Ivanov, M. M., **Philcox, O. H. E.**, Cabass, G., *et al.*, “Ultra-light axions and the S_8 tension: joint constraints from the cosmic microwave background and galaxy clustering”, *JCAP* **06** 023 (2023) ([arXiv](#)).
87. *Abdalla, E., Franco Abellán, G., Aboubrahim, A., Agnello, A., Akarsu, O., Akrami, Y., *et al.* (inc. **Philcox, O. H. E.**), “Cosmology intertwined: A review of the particle physics, astrophysics, and cosmology associated with the cosmological tensions and anomalies”, *JHEA* **34**, 49 – 211 (2022) ([arXiv](#)).
88. Villaescusa-Navarro, F., Genel, S., Anglés-Alcázar, D., Perez, L. A., Villanueva-Domingo, P., Wadekar, D., *et al.* (inc. **Philcox, O. H. E.**), “The CAMELS Project: Public Data Release”, *ApJS* **265**, 54 (2023) ([arXiv](#)).
89. Hou, J., Cahn, R. N., **Philcox, O. H. E.**, Slepian, Z., “Analytic Gaussian covariance matrices for galaxy N -point correlation functions”, *Phys. Rev. D* **106**, 043515 (2022) ([arXiv](#)).
90. Schmittfull, M., Simonović, M., Ivanov, M. M., **Philcox, O. H. E.**, Zaldarriaga, M., “Modeling Galaxies in Redshift Space at the Field Level”, *JCAP* **05** 059 (2021) ([arXiv](#)).
91. Villaescusa-Navarro, F., Anglés-Alcázar, D., Genel, S., Spergel, D. N., Somerville, R. S., Dave, R., *et al.* (inc. **Philcox, O. H. E.**), “The CAMELS project: Cosmology and Astrophysics with Machine Learning Simulations”, *ApJ* **915**, 71 (2021) ([arXiv](#)).
92. Wang, Y., Zhao, G.-B., Zhao, C., **Philcox, O. H. E.**, Alam, S., Tamone, A., *et al.*, “The clustering of the SDSS-IV extended Baryon Oscillation Spectroscopic Survey DR16 luminous red galaxy and emission line galaxy samples: cosmic distance and structure growth measurements using multiple tracers in configuration space”, *MNRAS* **498**, 3470 – 3483 (2020) ([arXiv](#)).
93. *Chudaykin, A., Ivanov, M. M., **Philcox, O. H. E.**, Simonović, M., “Nonlinear perturbation theory extension of the Boltzmann code CLASS”, *Phys. Rev. D* **102**, 063533 (2020) ([arXiv](#)).

Other Works

94. **Philcox, O. H. E.**, “An Introduction to the EFTofLSS”, *Lecture Notes*.

SELECTED RESEARCH TALKS

* = Virtual Talk

- 2026 Cosmic Rhythms, Rio de Janeiro, *Conference*
 New Frontiers in Cosmology, A Coruña, *Conference*
 National Taiwan University, *Physics Colloquium*
 *CosmoVerse Seminar, *Webinar* (Invited Talk)
 SLAC National Laboratory, *Colloquium*
 Max Planck-IAS-NTU Center Kick-Off, Princeton, *Conference* (Invited Talk)
 Laboratoire d'Annecy de Physique Théorique, Annecy, France *Seminar*
 CERN, Geneva, Switzerland *Seminar*
 New Synergies in Multiprobe Cosmology, KITP, Santa Barbara, *Conference*
- 2025 41st Institut d'Astrophysique de Paris Symposium, *Conference* (Invited Talk)
 Galileo Galilei Institute for Theoretical Physics, *Symposium* (Invited Plenary)
 21st Rencontres de Vietnam, *Conference* (Invited Talk)
 Understanding Cosmological Observations, Benasque, *Conference* (Invited Talk)
 GR24 & Amaldi16, *Conference* (Invited Plenary)
 Columbia University, *Physics Colloquium*
 Cosmological Probes of New Physics, Notre Dame, *Conference* (Invited Discussion)
 Brown University, *Theoretical Physics Seminar*
 Princeton University, *Cosmology Seminar*
 Particles vs. New Probes, Flatiron Institute, *Conference* (Invited Plenary)
 Boston University, *High-Energy Physics Seminar*
- 2024 21st Century Cosmology, Ashoka University, *Conference* (Invited Plenary)
 Cosmology in the Adriatic, Split, *Conference*
 University of Edinburgh, *Conference*
 Lawrence Berkeley National Laboratory, *Workshop* (Invited Plenary)
 University of Cambridge, *Astronomy Colloquium*
 Royal Society, London, *Discussion Meeting* (Invited Talk)
 Harvard University, *Cosmology Seminar*
 Massachusetts Institute of Technology, *Cosmology Seminar*
 58th Rencontres de Moriond, *Conference*
 Stanford University, *Physics Colloquium*
 Perimeter Institute for Theoretical Physics, *Colloquium*
- 2023 Taipei Institute of Astronomy & Astrophysics, *Conference* (Invited Talk)
 Max-Planck-Institute for Nuclear Physics, Heidelberg, *Particle Physics Seminar*
 ICTP Trieste, *Workshop*
 University of Montreal, *Astrophysics Seminar*
 *Parity-Violation from Home, *Conference*
 Ohio State University, *Cosmology and Astro-Particle Physics Seminar*
 *Early Universe / AliCPT Forum, University of Science and Technology of China, *Webinar*
 Sexten Center for Astrophysics, *Workshop* (Invited Plenary)
 Donostia International Physics Center, *Workshop* (Invited Talk)
 University of Pennsylvania, *PDT Partners Retreat* (Invited Talk)
 Yukawa Institute for Theoretical Physics, Kyoto, *Conference* (Invited Talk)
 Kavli IPMU, Tokyo, *Astronomy Seminar*
 Stony Brook, *Cosmology Seminar*
 Cosmology on Safari, *Conference*
 Johns Hopkins University, *Particle Physics Seminar*
 University of Maryland, *Particle Physics Seminar*
 *Copernicus Series, *Cosmology Webinar*
 *University of Oxford, *Cosmology Seminar*

OUTREACH TALKS

- 2026 *Astronomy on Tap*, San Francisco (Public Talk)
 Discover Our Universe, Kavli Institute for Particle Astrophysics & Cosmology (Public Lecture)
- 2025 *Stanford University Undergraduate Physics Club* (Faculty Lunch Talk)
 Taste of Science, New York (Public Lecture)
- 2023 *Into The Unknown* (Public Podcast)

TEACHING & LECTURING

- 2026 Cosmology, *Stanford University* (Physics 361)
 Croucher Advanced Study Institute on Primordial Non-Gaussianity, Hong Kong (Videos: [Part I](#), [Part II](#))
- 2024 Essential Cosmology for the Next Generation IX, Mexico, *Winter School* (Invited Lecture Series)

MEDIA

1. “Gravity follows Newton and Einstein’s rules, even at cosmic scales”, [Simons Foundation](#), 15 Apr. 2026.
2. “Scientists detect mysterious suppression in cosmic structure growth”, [Phys.org](#), 3 Jan. 2025.
3. “Universe’s ‘Cosmological Collider’ Lands 3 Scientists \$100,000 Physics Prize”, [Live Science](#), 14 Sep. 2023.
4. “What Happened Right After the Universe Began?”, [Simons Foundation](#), 30 Aug. 2023.
5. “The Cosmos as a Colloid”, [Physics Magazine](#), 14 Mar. 2023.
6. “Pinpoint Simulations Provide Perspective on Universe Structure”, [IAS News](#) & [Phys.org](#), 14 Mar. 2023.
7. “Spatial Patterns In Distribution of Galaxies”, [Princeton News](#) & [ScienceDaily](#), 14 Mar. 2023.
8. “Is the Universe Asymmetrical?”, [Columbia News](#), 27 Feb. 2023.
9. “Do We Live in a Mirror Universe?”, [Into The Unknown Podcast](#), 26 Jan. 2023.
10. “Asymmetry Detected in the Distribution of Galaxies”, [Quanta](#), 5 Dec. 2022.
11. “The Universe is Surprisingly Lopsided and We Don’t Know Why”, [New Scientist](#), 18 Jun. 2022.

CURRENT GRANTS

- 2026 NSF PHY-2609396 (PI; \$140 000): *Probing High-Energy Physics with the Galaxy Bispectrum*

PRIZES

- 2024 New Horizons in Physics Prize, *Breakthrough Prize Foundation*
- 2023 Buchalter Cosmology Prize (First Prize)
- 2022 Simons Society of Fellows (Junior Fellowship)
 NHFP Einstein Fellowship, *declined*
 LBL Chamberlain Fellowship, *declined*
 Cambridge Kavli Fellowship, *declined*
- 2018 Herchel-Smith Scholarship, *Cambridge → Harvard*
 Institute of Astronomy Prize, *Cambridge*
- 2017 Holgate Pollard Memorial Prize, *Cambridge*

PROFESSIONAL ACTIVITIES

Referee	MNRAS (2020–), JCAP (2020–), MPLA (2021–), Phys. Rev. Lett. (2022–) Phys. Rev. D (2022–), ApJS (2022–), Phys. Dark Univ. (2023–)
Conferences	Large-Scale Parity Violation (Taiwan 2023, SOC)
Review Panels	NASA Astrophysics Theory Program (2023)
Mentor	<i>Dhayaa Anbajagane</i> NASA Hubble Fellow (2026–) <i>Beatriz Tucci</i> Stanford Postdoctoral Fellow (2025–)
Advisor	<i>Cooper Jacobus</i> Stanford Graduate Student (2026–) <i>Sixiang Yang</i> Stanford Graduate Student (2025–) <i>Hanyue Wang</i> Stanford Graduate Student (2026, rotation) <i>Alessandro Russo</i> Stanford Graduate Student (2025, rotation) <i>Anita Dunsmore</i> Stanford Graduate Student (2025, rotation) <i>Sam Goldstein</i> Columbia Graduate Student (2022–2025, various projects) <i>John Moynihan</i> Columbia Undergraduate Student (2023, summer project) <i>Jess Boyland</i> Simons-NSBP Undergraduate Scholar (2020–2021, summer project)
Coadvisor	<i>Luca Abu El-Haj</i> Columbia Undergraduate Student (2025, summer project) <i>Gemma Zhang</i> Harvard Graduate Student (2023–2025) <i>Kristen Surrao</i> Columbia Graduate Student (2022–2023) <i>Gerrit Farren</i> Cambridge Graduate Student (2020–2022)

MISCELLANEOUS

Computing Languages	PYTHON, C++, JULIA, MATHEMATICA, CUDA
Codes Developed	POLYSPEC, POLYBIN3D, ENCORE, NPCFs.jl, CLASS-PT, SPECTRA-WITHOUT-WINDOWS, EFFECTIVEHALOS, HIPSTER, RASCALC
Teaching	6 years of online tutoring (high-school to Masters level) Teaching assistant for Princeton introductory astronomy class (AST203) TEFL qualification in English teaching
Other	DipABRSM in Music Performance (Distinction)

REFERENCES

Prof. J C Hill
(Postdoctoral Mentor)
Columbia University
914 Pupin Hall
New York, NY 10027
Tel: +1 (212) 854-7815
jch2200@columbia.edu

Prof. D N Spergel
(PhD Advisor)
Simons Foundation
160 5th Ave.
New York, NY 10010
Tel: +1 (609) 258-3589
dspergel@simonsfoundation.org

Prof. M Zaldarriaga
(PhD Advisor)
Institute for Advanced Study
1 Einstein Drive
Princeton, NJ 08540
Tel: +1 (609) 734-8058
matiasz@ias.edu